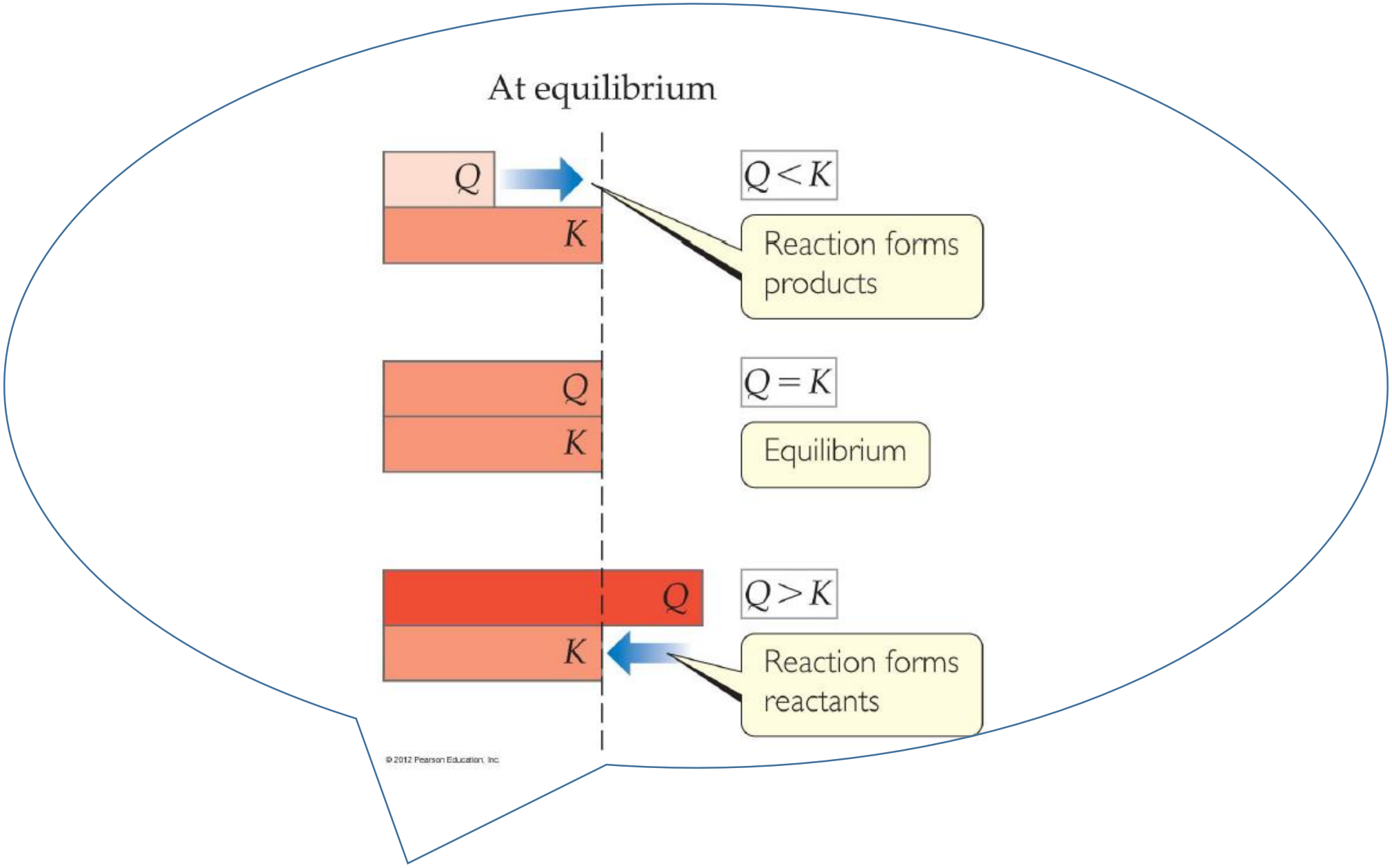
EXERCISES ABOUT CHAPTER 15 Chemical Equilibrium – Chapter 16 Acid-Base Equilibria



* ~~Initial~~ concentrations are given for the following reaction

$$Ag^{+}_{(aq)} + Fe^{2+}_{(aq)} \rightleftharpoons Ag_{(s)} + Fe^{3+}_{(aq)}$$

$[Ag^{+}]_0 = 0.20\ M$

$[Fe^{2+}]_0 = 0.10\ M$

$[Fe^{3+}]_0 = 0.30\ M$

$K_c = 2.98$

When the reaction reached the equilibrium What are the concentrations of ions ?

$$Q_c = \frac{[Fe^{3+}]}{[Ag^+][Fe^{2+}]} = \frac{(0.30)}{(0.20)\cdot(0.10)} = 15.0$$

$Q_c > K_c \implies$ reaction forms reactants

$\leftarrow$ (reaction continue to reactants)

$$Ag^+ + Fe^{2+} \rightleftharpoons Ag(s) + Fe^{3+}$$

	0.2 M	0.1 M	0.3 M
	+x	+x	−x
Eq.	(0.2 + x) M	(0.1 + x) M	(0.3 − x) M

$$K_c = \frac{(0.3 - x)}{(0.2 + x)(0.1 + x)} = 2.98 \qquad x = 0.11$$

$[Ag^{+}]_e = 0.31\ M$

$[Fe^{2+}]_e = 0.21\ M$

$[Fe^{3+}]_e = 0.19\ M = (0.3 - 0.11)$

At 448 °C, the equilibrium constant K_c for the reaction

$$H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$$

is 50.5. Predict in which direction the reaction proceeds to reach equilibrium if we start with 2.0×10^{-2} mol of HI, 1.0×10^{-2} mol of H_2, and 3.0×10^{-2} mol of I_2 in a 2.00-L container.

Solve The initial concentrations are

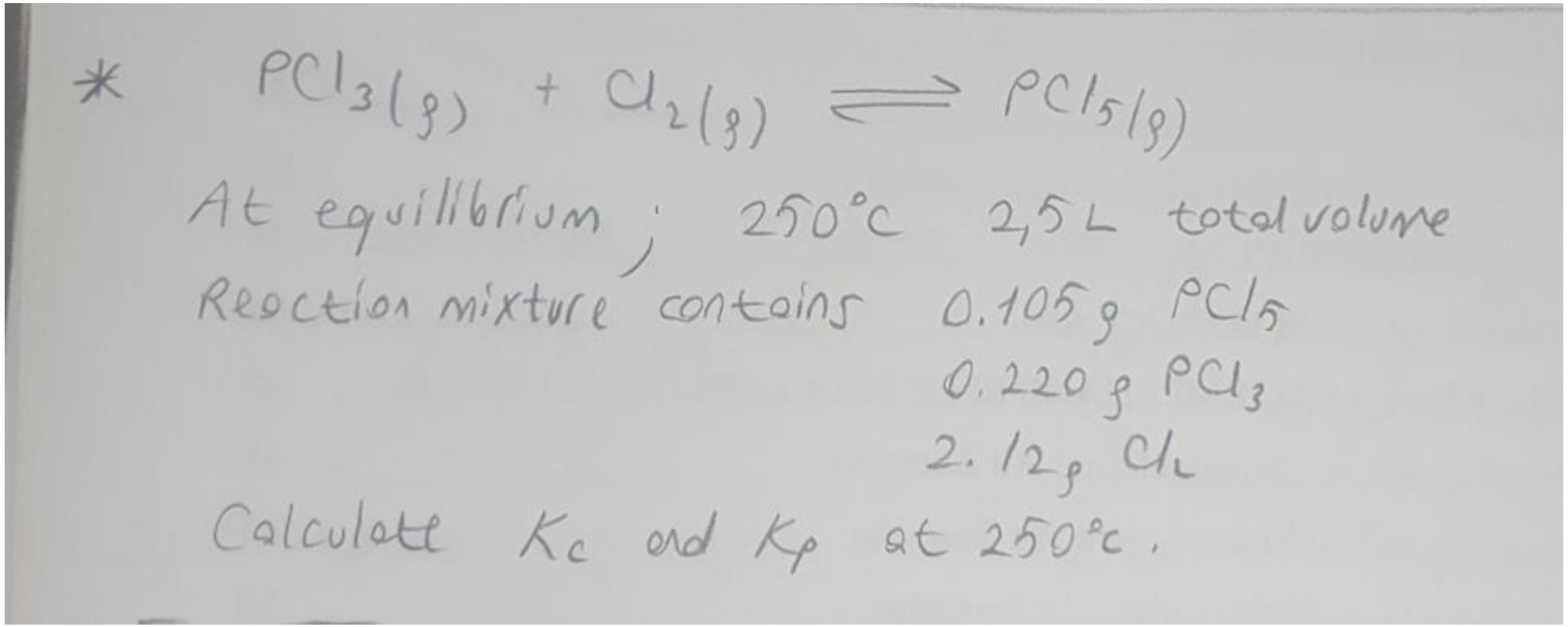
$$[HI] = 2.0 \times 10^{-2}\,\text{mol}/2.00\,\text{L} = 1.0 \times 10^{-2}\,M$$
$$[H_2] = 1.0 \times 10^{-2}\,\text{mol}/2.00\,\text{L} = 5.0 \times 10^{-3}\,M$$
$$[I_2] = 3.0 \times 10^{-2}\,\text{mol}/2.00\,\text{L} = 1.5 \times 10^{-2}\,M$$

The reaction quotient is therefore

$$Q_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{(1.0 \times 10^{-2})^2}{(5.0 \times 10^{-3})(1.5 \times 10^{-2})} = 1.3$$

Because $Q_c < K_c$, the concentration of HI must increase and the concentrations of H_2 and I_2 must decrease to reach equilibrium; the reaction as written proceeds left to right to attain equilibrium.

$$K_p = K_c (RT)^{\Delta n}$$



$$K_c = \frac{[PCl_5]}{[PCl_3] \times [Cl_2]} = \frac{2.017 \times 10^{-4}}{(6.41 \times 10^{-4}) \times (0.012)} = 26.2$$

$$[PCl_5] = \frac{0.105\ g}{208.2\ g/mol \times 2.50\ L} = 2.017 \times 10^{-4}\ M$$

$$[PCl_3] = \frac{0.229\ g}{137.3\ g/mol \times 2.50\ L} = 6.41 \times 10^{-4}\ M$$

$$[Cl_2] = \frac{2.12\ g}{70.9\ g/mol \times 2.5\ L} = 0.012\ M$$

$$K_p = K_c \cdot (RT)^{\Delta n} = 26.2\ (0.082 \times 523)^{-1} = 0.613$$

* Formamide decomposition reaction

$HCONH_2(g) \rightleftharpoons NH_3(g) + CO(g)$ $K_c = 4.84$ @ 400 K

If 0.186 mole $HCONH_2(g)$ reacted in a 2.16 L volume flask at 400 K, Calculate the total pressure at equilibrium.

initial conc. $[HCONH_2]_0 = \frac{0.186 \text{ mole}}{2.16 \text{ L}} = 0.0861 \text{ M}$

	$HCONH_2(g)$	$\rightleftharpoons$	$NH_3(g)$	+	$CO(g)$
	0.0861 M		0		0
	$-x$		$+x$		$+x$
Eq.	$(0.0861-x)$ M		x M		x M

$$K_c = \frac{[NH_3]\cdot[CO]}{[HCONH_2]} = \frac{x \cdot x}{(0.0861-x)} = 4.84$$

$$x^2 = 0.417 - 4.84x$$

$$x^2 + 4.84x - 0.417 = 0 \longrightarrow x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$\frac{-4.84 \pm \sqrt{23.4-(-1.67)}}{2} \Rightarrow$$

$x = 0.08$ M

$x = -4.92$ M

total concentration at equilibrium

$[HCONH_2]_e = 6.1 \times 10^{-3}\ M \quad (0.0861 - x)$

$[NH_3]_e = 0.08\ M \quad (x)$

$[CO]_e = 0.08\ M \quad (x)$

$0.17\ M$

$P_{total} = 0.17\ M \times 0.082 \frac{L\,atm}{mol\,K} \times 400\ K = \boxed{5.58\ atm}$

Örnek: 0.1 M CH_3COOH solution pH is measured as pH = 2.8
Calculate.

	CH_3COOH	+ H_2O ⇌	H_3O^+	+ CH_3COO^-	$K_a = 1.8 \times 10^{-5}$
initial	0.1		0	0	
change	$-x$		$+x$	$+x$	
	$0.1 - x$		x	x	

$$K_a = 1.8 \times 10^{-5} = \frac{x^2}{0.1 - x}$$

$0.1 - x \cong 0.1$ kabul

$x = 1.3 \times 10^{-3} M = 0.0013 M$

Check the assumption: $0.1 - 0.0013 = 0.099$ %1'i

$pH = -\log[H_3O^+] = 2.89$

örnek: Calculate pH for 0.0025 M $CH_3NH_2(aq)$ solution.

$K_b = 4.2 \times 10^{-4}$

	CH_3NH_2	+ H_2O	⇌	$CH_3NH_3^+$	+ OH^-
initial	0.0025			0	0
change	−x			+x	+x
eq.	(0.0025−x)			x	x

$$K_b = 4.2 \times 10^{-4} = \frac{x^2}{0.0025 - x}$$

$0.0025 - x \cong 0.0025$ kabul

$x = 1 \times 10^{-3}\ M = 0.001\ m$

check the assumption $0.0025 - 0.001$

→ %40'

you cannot neglect x value

you should solve

$$\frac{x^2}{0.0025 - x} = 4.2 \times 10^{-4}$$

$$x^2 + (4.2 \times 10^{-4})x - 1.1 \times 10^{-6} = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x = 8.4 \times 10^{-4}\ M = [OH^-]$

$pOH = 3.08$

$pH = 10,92$

Strong Acids - Strong Bases - pH

* To prepare 12.5 L, pH = 2.1 solution

How many mL $HCl_{(aq)}$ needed

↳ 36 % (w/w) $d = 1.18\ g/mL$ $HCl: 36.5\ g/mol$

Solution :

$$\underbrace{HCl}_{10^{-2.1}\ M} + H_2O \rightleftharpoons Cl^- + \underbrace{H_3O^+}_{10^{-2.1}\ M}$$

istiyor

$$\frac{10^{-2.1}\ mol}{1\ L} \times 12.5\ L = 0.0993\ mol\ HCl\ needed$$

$$0.0993\ mol \times \frac{36.5\ g}{mol} \times \frac{100}{36} \times \frac{1\ mL}{1.18\ g} = \boxed{8.53\ mL}$$

* Calculate $[H_3O^+]$ and pH for a saturated $Ba(OH)_2(aq)$ solution that contains 3.9 g $Ba(OH)_2 \cdot 8H_2O$ in 100 mL. ($Ba(OH)_2 \cdot 8H_2O$; 315.5 g/mol)

Solution :

$$[OH^-] = \frac{3.9\ g}{100\ mL} \times \frac{1000\ mL}{1\ L} \times \frac{1\ mol\ Ba(OH)_2 \cdot 8H_2O}{315.5\ g} \times \frac{2\ mol\ OH^-}{1\ mol\ Ba(OH)_2 \cdot 8H_2O} = 0.25\ M$$

$$[H_3O^+] = \frac{Kw}{[OH^-]} = \frac{1 \times 10^{-14}}{0.25\ M} = 4 \times 10^{-14}\ M \qquad pH = -\log\left(4 \times 10^{-4}\right) = 13.40$$

Watch This!

Because

$$[H_3O^+]\,[OH^-] = K_w = 1.0 \times 10^{-14}$$

we know that

$$-\log [H_3O^+] + -\log [OH^-] = -\log K_w = 14.00$$

or, in other words,

$$pH + pOH = pK_w = 14.00$$

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Weak Acids - Weak Bases - pH

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Ionization reaction of phenilacetic acid is given

$$HC_8H_7O_2(aq) + H_2O(\ell) \rightleftharpoons H_3O^+_{(aq)} + C_8H_7O_2^-(aq)$$

$$K_a = 4.9 \times 10^{-5}$$

a) What is the $[C_8H_7O_2^-]$ for 0,186 M $HC_8H_7O_2$ solution?

b) What is the pH value for 0,121 M $HC_8H_7O_2$ solution?

solution :

a) $HC_8H_7O_2(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + C_8H_7O_2^-(aq)$

	$HC_8H_7O_2(aq)$	$H_2O(l)$	$H_3O^+(aq)$	$C_8H_7O_2^-(aq)$
initial :	0.186 M	0	0	0
change :	$-x$		$+x$	$+x$
eq.	$(0.186-x)$ M		x M	x M

$$K_a = 4.9 \times 10^{-5} = \frac{x \cdot x}{0.186-x} \cong \frac{x^2}{0.186} \qquad x \ll 0.186 \checkmark$$

$$x = \sqrt{4.9 \times 10^{-5} \times 0.186} = \underbrace{0.003 \text{ M}}_{\text{0.186 nın \%5 inden azdır.}} = [C_8H_7O_2^-] = [H_3O^+]$$

6) $HC_8H_7O_2(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + C_8H_7O_2^-(aq)$

	$HC_8H_7O_2(aq)$	$H_2O(l)$	$H_3O^+(aq)$	$C_8H_7O_2^-(aq)$
initial:	0.121 M	0	≈ 0	0
change	$-x$		$+x$	$+x$
eq.	$(0.121-x)$ M		x M	x M

$$K_a = 4.9 \times 10^{-5} \cong \frac{x^2}{0.121}$$

$x = 0.0024$ M

$[H_3O^+] = 0.0024$

$$pH = -\log(0.0024) = 2.62$$

* Calculate the mass of benzoic acid C_6H_5COOH, that should be dissolved in 350 mL water to prepare a solution pH = 2.85.

$$C_6H_5COOH + H_2O \rightleftharpoons H_3O^+ + C_6H_5COO^- \qquad K_a = 6.3 \times 10^{-5}$$

solution : $[H_3O^+] = 10^{-2.85} = 1.4 \times 10^{-3}\ M$

we need the initial molarity of acid.

	C_6H_5COOH	$+ H_2O \rightleftharpoons$	H_3O^+	$+ C_6H_5COO^-$
initial	S		0	0
change	-0.0014		$+0.0014$	$+0.0014$
eq	$S - 0.0014$		0.0014	0.0014

$$K_a = \frac{[H_3O^+]\cdot[C_6H_5COO^-]}{[C_6H_5COOH]} = \frac{(0.0014)^2}{S - 0.0014} = 6.3 \times 10^{-5}$$

$$S - 0.0014 = 0.031 \Rightarrow S = 0.0324\ m$$

$$\frac{0.0324\ mol}{1\ L} \times \frac{1\ L}{1000\ mL} \times 350\ mL \times \frac{122\ g}{1\ mol} = \boxed{1.4\ g\ C_6H_5COOH}$$

At 25°C, 275 mL 1-propylamine vapor is taken and dissolved in 0.5 L H_2O.

For 1-propylamine $pK_b = 3.43$ and vapor pressure 316 Torr.

a) Calculate pH for aqueous solution

b) For the same pH how many mg NaOH should be dissolved in 0.5 L H_2O ?

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solution:

a) $n_{propylamine} = \frac{P.V}{R.T} = \frac{\left(316\ Torr \times \frac{1\ atm}{769\ torr}\right) \times 0.275\ L}{0.082\ \frac{L \cdot atm}{mol \cdot K} \times 298\ K} = 4.67 \times 10^{-3}\ mol$

$[propylamine] = \frac{4.67 \times 10^{-3}\ mol}{0.5\ L} = 9.35 \times 10^{-3}\ M$

$-\log K_b = 3.43$

$K_b = 10^{-3.43} = 3.7 \times 10^{-4}$

	$CH_3CH_2CH_2NH_2(aq)$	+ $H_2O(l)$	$\rightleftharpoons$	$CH_3CH_2CH_2NH_3^+(aq)$	+ $OH^-(aq)$
Initial	9.35×10^{-3}			0	~0
change	$-x$			$+x$	$+x$
eq.	$(9.35\times10^{-3}-x)$ m			x	x

$$K_b = 3.7\times10^{-4} = \frac{x^2}{9.35\times10^{-3}-x}$$

$$x^2 + 3.7\times10^{-4}x - 3.5\times10^{-6} = 0 \qquad x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$x = \frac{-3.7\times10^{-4} \pm \sqrt{(3.7\times10^{-4})^2 - 4\cdot(1)\cdot(-3.5\times10^{-6})}}{2\cdot(1)} \qquad x = 1.695\times10^{-3}\ M = [OH^-]$$

$$pOH = -\log[OH^-] = 2.77$$

$$\boxed{pH = 11.23}$$

$$[OH^-] = 1.7 \times 10^{-3}\ M$$

$$1,7 \times 10^{-3}\ M = \frac{n\ mol\ NaOH}{0,5\ L} \Rightarrow n = 8,5 \times 10^{-4}\ mol\ NaOH$$

$$8.5 \times 10^{-4}\ mol \times 40\ g/mol = 0.034\ g$$

$$\boxed{34\ mg\ NaOH}$$

The solubility of 1-naphtylamine, $C_{10}H_7NH_2$, is given as 1 g per 590 g of water. What is the approximate pH of a saturated aqueous solution of 1-naphtylamine?
(1-naphtylamine $mw = 143.2$ g/mol, water $d = 1$ g/mL)

$$C_{10}H_7NH_2 + H_2O \rightleftharpoons C_{10}H_7NH_3^+ + OH^- \quad pK_b = 3.92$$

$$[C_{10}H_7NH_2] = \frac{1\text{ g } C_{10}H_7NH_2}{590\text{ g } H_2O} \times \frac{1\text{ g } H_2O}{1\text{ mL } H_2O} \times \frac{1\text{ mL}}{1000\text{ L}} \times \frac{1\text{ mol } C_{10}H_7NH_2}{143.2\text{ g } C_{10}H_7NH_2}$$

$$[C_{10}H_7NH_2] = 0.012\text{ M}$$

$$K_b = 10^{-pK_b} = 10^{-3.92} = 1.2 \times 10^{-4}$$

$$C_{10}H_7NH_2 + H_2O \rightleftharpoons C_{10}H_7NH_3^+ + OH^-$$

	$C_{10}H_7NH_2$	H_2O	$C_{10}H_7NH_3^+$	OH^-
initial	0.012 M		0	0
change	$-x$		$+x$	$+x$
equilibrium	$(0.012-x)$		x	x

$$K_b = \frac{x^2}{(0.012-x)} = 1.2\times10^{-4} \qquad x \ll 0.012 \Rightarrow \frac{x^2}{0.012} = 1.2\times10^{-4}$$

$$\rightarrow x = 0.0012\ M$$

$$[OH^-] = 0.0012\ M$$

$$pOH = -\log 0.0012 = 2.92$$

$$pH = 14 - 2.92 = 11.08$$